

Akshat Anand

Software Engineer | Full-Stack | AI Platforms | Real-Time & 3D

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Summary

Software engineer, 3.5 years, end-to-end ownership across frontend, backend, AI and real-time 3D. Ships production systems: payment and subscription flows, guarded multi-provider LLM pipelines, browser-based 3D avatar tooling, and a real-time lip-sync engine written from scratch against raw audio.

Technical Skills

- **Languages/Frontend:** TypeScript, JavaScript, React, Next.js, Redux, Zustand, Tailwind CSS, GSAP.
- **Backend:** Node.js, Express.js, PostgreSQL, MongoDB, Prisma, REST APIs, JWT auth, webhooks, payment gateway integration, python.
- **AI:** LLMs, LangChain, RAG, OpenAI GPT-4o, Google Gemini, Pinecone, embeddings, TTS, LLM guardrails, multi-provider fallback.
- **3D & Real-Time:** Three.js, WebGL rendering pipelines, morph-target animation, Web Audio API, WebSocket, Socket.IO, event-driven streaming.
- **DevOps:** Docker, GitHub Actions CI/CD, React Native, Expo, Capacitor, Tauri.

Experience

Software Engineer, Metabrix Lab | Aug 2025 – Present

- **Real-time lip-sync engine, built from scratch:** drives character mouth shapes directly from live audio waveform analysis frame by frame, with no pre-baked phoneme data and no third-party lip-sync SDK.
- **3D avatar configurator:** runtime customisation of a live Three.js avatar: skin and hair colour, material swaps, environment lighting, animation clip switching, applied without rebuilding the scene.
- **Payments end to end:** integrated the checkout and subscription flow powering paid access, with server-side verification and webhook-driven entitlement state across renewals, failures, and retries.
- **Guardrails on the production AI pipeline:** input/output filtering against prompt injection and unsafe generations, plus multi-provider fallback (OpenAI, Gemini) so a provider outage degrades instead of breaking a session.
- **Multi-provider LLM + TTS streaming:** token-level streaming into production dashboards over an event-driven Node.js pipeline with per-provider latency instrumentation, integrated via LangChain.
- **Platform delivery,** replaced monolithic polling with event-driven services (~40% faster dashboard load, 3 dashboards); GitHub Actions CI/CD with test gates and image versioning for zero-downtime deploys across 10+ releases.

Freelance Software Engineer, Independent | Mar 2025 – Jul 2025

Frontend Developer, Zimension 3D Technologies | Mar 2023 – Dec 2024

- **FPS improvement** in browser-based 3D scenes, redesigned the Three.js rendering pipeline with geometry instancing, level-of-detail switching, and texture atlasing (verified via Chrome DevTools).
- **Reduced re-renders:** refactored state to Redux with memoised selectors (React Profiler), improving responsiveness on low-powered devices.
- **3 high-severity findings closed to zero** in an internal security audit overhauled access control and input sanitisation across the application.

Projects

Requestceptor: [Network Interception & API Security Suite](#)*: on going 7-services Node.js monorepo that intercepts, mutates, and replays HTTP traffic through a MITM proxy, a declarative match/action rule engine, a real-time WebSocket traffic console with custom filter grammar, request replay, fuzzing, load testing, and an 11-module vulnerability scanner (SQLi, XSS, SSRF, IDOR, CORS, CSP, auth, TLS).

Portify: [Multi-platform Web-to-App Pipeline](#): Architected and built a cloud service and CLI pipeline that converts baseline web builds into native Android (APK), and Windows (EXE) applications via Tauri v2. Incremental fast-swap builds minimizing Rust compile times. Full toolchain auto-setup script for Linux, Windows (Rust, JDK, Android SDK/NDK). Isolated concurrent build queue and an HTTP API for uploading zip payloads and tracking build completion.

Numerologist AI: [Cross-Platform AI Product \(Web + Android\)](#): single Next.js codebase shipped to web and native via Capacitor, behind a hardened Express OpenAI-proxy keeping the API key server-side.

Education & Certifications

BCA, CS — L.N.M University

Full Stack Fellowship — Crio.do (2023)

GenAI Cohort — Chai aur Code (2025)